

Orthogonal Vectors

$$\bar{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \quad \bar{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

$$\bar{u} \cdot \bar{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

DEFINITION

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are orthogonal (to each other) if $\mathbf{u} \cdot \mathbf{v} = 0$.

Observe that the zero vector is orthogonal to every vector in $\mathbb{R}^n$ because $\mathbf{0}^T \mathbf{v} = 0$ for all $\mathbf{v}$.

The next theorem provides a useful fact about orthogonal vectors. The proof follows immediately from the calculation in (1) above and the definition of orthogonality. The right triangle shown in Figure 6 provides a visualization of the lengths that appear in the theorem.

$$\bar{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\bar{v} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$\bar{u} \cdot \bar{v} = 3 \cdot (-2) + 2 \cdot 3 = 0$$

$$\bar{u} \perp \bar{v} \quad (\text{perpendicular})$$

$$\bar{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\bar{v} = \begin{bmatrix} 7 \\ -3 \\ -4 \end{bmatrix}$$

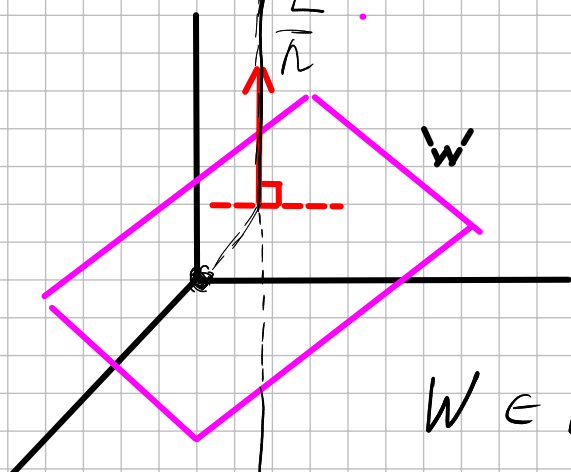
$$\bar{u} \cdot \bar{v} = 1 \cdot 7 + 1 \cdot (-3) + 1 \cdot (-4) = 0$$

$$\bar{u} \perp \bar{v}$$

Orthogonal Complements

To provide practice using inner products, we introduce a concept here that will be of use in Section 6.3 and elsewhere in the chapter. If a vector $\mathbf{z}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then $\mathbf{z}$ is said to be orthogonal to W . The set of all vectors $\mathbf{z}$ that are orthogonal to W is called the orthogonal complement of W and is denoted by $W^\perp$ (and read as " W perpendicular" or simply " W perp").

Set of all " $\mathbf{z}$ " are is called ort/compl. of W
 $W^\perp$ - orthogonal complement



$$\mathbf{0} \in L$$

$$\mathbf{0} \in W$$

$$W^\perp = L$$

$$L^\perp = W$$

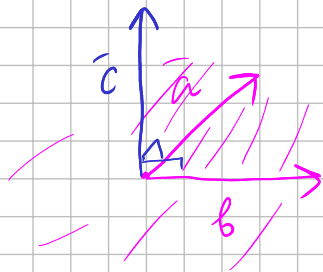
$$W \in \mathbb{R}^3$$

$$L \in \mathbb{R}^3$$

W, L are subspaces of $\mathbb{R}^3$

Properties

- 1° $\bar{x} \in W^\perp$ if & ($\Leftrightarrow$) $\bar{x} \perp$ to every vector from the spanning set



$$\left. \begin{array}{l} \bar{c} \perp \bar{a} \\ \bar{c} \perp \bar{b} \end{array} \right\} \bar{c} = \text{span}\{\bar{a}, \bar{b}\}^\perp$$

- 2° if W is a subspace of R^n then $W^\perp$ is also a subspace of R^n

- 3° if $A \in R_{m \times n}$, $\text{Row } A$, $\text{Col } A$, $\text{Nul } A$

$$(\text{Row } A)^\perp = \text{Nul } A$$

$$(\text{Col } A)^\perp = \text{Nul } (A^T)$$

vectors of $\text{Row } A \perp$ vectors of $\text{Nul } A$

$$\bar{a} \in \text{Row } A$$

$$\bar{b} \in \text{Nul } A$$

$$\bar{a} \cdot \bar{b} = 0$$

$$\bar{c} \in \text{Col } A,$$

$$\bar{c} \perp \bar{x} \in \text{Nul } (A^T)$$

6.2 ORTHOGONAL SETS

A set of vectors $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ is said to be an **orthogonal set** if each pair of distinct vectors from the set is orthogonal, that is, if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$.

EXAMPLE 1 Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set, where

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

$$\bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_2 = 3 \cdot (-1) + 1 \cdot 2 + 1 \cdot 1 = -3 + 2 + 1 = 0 \quad \checkmark$$

$$\begin{aligned} \bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_3 &= 3 \cdot (-1/2) + 1 \cdot (-2) + 1 \cdot 7/2 = \\ &= -\frac{3}{2} - 2 + \frac{7}{2} = 2 - 2 = 0 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \bar{\mathbf{u}}_2 \cdot \bar{\mathbf{u}}_3 &= -1 \cdot (-1/2) + 2 \cdot (-2) + 1 \cdot 7/2 = \\ &= 1/2 - 4 + 7/2 = 4 - 4 = 0 \quad \checkmark \end{aligned}$$

THEOREM 4

If $S = \{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal set of **nonzero** vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S .

Proof: Assume that **the set S is lin/dep**

$\Rightarrow \exists c_1 \dots c_p$ (not all zeros): False

$$(*) \quad c_1 \cdot \bar{\mathbf{u}}_1 + c_2 \bar{\mathbf{u}}_2 + \dots + c_p \cdot \bar{\mathbf{u}}_p = \mathbf{0}$$

$$(*) \cdot \bar{\mathbf{u}}_1 \quad (\bar{\mathbf{u}}_1 \neq \mathbf{0})$$

$$c_1 \cdot \bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_1 + c_2 \cdot \bar{\mathbf{u}}_2 \cdot \bar{\mathbf{u}}_1 + \dots + c_p \cdot \bar{\mathbf{u}}_p \cdot \bar{\mathbf{u}}_1 = 0 \cdot \bar{\mathbf{u}}_1$$

$$c_1 \cdot \bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_1 = 0 \Rightarrow c_1 = 0$$

$$(*) \cdot \bar{\mathbf{u}}_2 \quad (\bar{\mathbf{u}}_2 \neq \mathbf{0}) \quad c_1 \cdot \bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_2 + c_2 \bar{\mathbf{u}}_2 \cdot \bar{\mathbf{u}}_2 + \dots + c_p \bar{\mathbf{u}}_p \cdot \bar{\mathbf{u}}_2 = 0$$

$$c_2 = 0 \dots c_p = 0 \quad \Downarrow \Rightarrow S - \text{lin indep} \quad \blacktriangle$$

DEFINITION

An orthogonal basis for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

The next theorem suggests why an orthogonal basis is much nicer than other bases. The weights in a linear combination can be computed easily.

THEOREM 5

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y}$ in W , the weights in the linear combination

$$\mathbf{y} = c_1 \mathbf{u}_1 + \dots + c_p \mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j} \quad (j = 1, \dots, p)$$

$\mathbb{R}^2$

B_1 basis $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

B_2 basis $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right\}$

B_1 - orth. basis

B_2 is not orth.

EXAMPLE 2 The set $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ in Example 1 is an orthogonal basis for $\mathbb{R}^3$.

Express the vector $\mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as a linear combination of the vectors in S .

$$\mathbf{u}_1 = (3, 1, 1)$$

$$\mathbf{u}_2 = (-1, 2, 1)$$

$$\mathbf{u}_3 = (-1/2, -2, 7/2)$$

S is an orth. set.

$$\mathbf{y} = c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + c_3 \mathbf{u}_3$$

$$c_1 = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}$$

$$\mathbf{y} \cdot \mathbf{u}_1 = 6 \cdot 3 + 1 \cdot 1 - 8 \cdot 1 = 18 + 1 - 8 = 11$$
$$\mathbf{u}_1 \cdot \mathbf{u}_1 = 3^2 + 1^2 + 1^2 = 11$$

$$c_1 = \frac{11}{11} = 1$$

$$c_2 = \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}$$

$$\mathbf{y} \cdot \mathbf{u}_2 = 6 \cdot (-1) + 1 \cdot 2 - 8 \cdot 1 = -6 + 2 - 8 = -12$$
$$\mathbf{u}_2 \cdot \mathbf{u}_2 = (-1)^2 + 2^2 + 1^2 = 6$$

$$c_2 = \frac{-12}{6} = -2$$

$$c_3 = \frac{\mathbf{y} \cdot \mathbf{u}_3}{\mathbf{u}_3 \cdot \mathbf{u}_3}$$

$$\mathbf{y} \cdot \mathbf{u}_3 = 6 \cdot (-1/2) + 1 \cdot (-2) - 8 \cdot (7/2) =$$
$$= -3 - 2 - 28 = -33$$
$$\mathbf{u}_3 \cdot \mathbf{u}_3 = (-1/2)^2 + (-2)^2 + (7/2)^2 =$$

$$= \frac{1}{4} + 4 + \frac{49}{4} = \frac{50}{4} + 4 = 14 + \frac{10}{4} = 16 \frac{1}{2}$$

$$c_3 = \frac{-33}{16.5} = -2$$

$$\mathbf{y} = 1 \cdot \mathbf{u}_1 - 2 \cdot \mathbf{u}_2 - 2 \cdot \mathbf{u}_3$$

Orthonormal Sets

A set $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an **orthonormal set** if it is an **orthogonal set** of **unit** vectors. If W is the subspace spanned by such a set, then $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an **orthonormal basis** for W , since the set is automatically linearly independent, by Theorem 4.

The simplest example of an orthonormal set is the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ for $\mathbb{R}^n$. Any nonempty subset of $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is orthonormal, too. Here is a more complicated example.

EXAMPLE 5 Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix}$$

1) $\|\bar{\mathbf{v}}_1\| = 1, \quad \|\bar{\mathbf{v}}_2\| = 1, \quad \|\bar{\mathbf{v}}_3\| = 1$

2) $\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_2 = 0, \quad \bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_3 = 0, \quad \bar{\mathbf{v}}_2 \cdot \bar{\mathbf{v}}_3 = 0$

1) $\|\bar{\mathbf{v}}_1\| = \sqrt{\left(\frac{3}{\sqrt{11}}\right)^2 + \left(\frac{1}{\sqrt{11}}\right)^2 + \left(\frac{1}{\sqrt{11}}\right)^2} = \sqrt{\frac{9}{11} + \frac{1}{11} + \frac{1}{11}} = 1$

$$\|\bar{\mathbf{v}}_2\| = \sqrt{\left(-\frac{1}{\sqrt{6}}\right)^2 + \left(\frac{2}{\sqrt{6}}\right)^2 + \left(\frac{1}{\sqrt{6}}\right)^2} = \sqrt{\frac{1}{6} + \frac{4}{6} + \frac{1}{6}} = 1$$

$$\|\bar{\mathbf{v}}_3\| = \sqrt{\left(-\frac{1}{\sqrt{66}}\right)^2 + \left(-\frac{4}{\sqrt{66}}\right)^2 + \left(\frac{7}{\sqrt{66}}\right)^2} = \sqrt{\frac{1}{66} + \frac{16}{66} + \frac{49}{66}} = 1$$

2) $\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_2 = \frac{3}{\sqrt{11}} \cdot \left(-\frac{1}{\sqrt{6}}\right) + \frac{1}{\sqrt{11}} \cdot \frac{2}{\sqrt{6}} + \frac{1}{\sqrt{11}} \cdot \frac{1}{\sqrt{6}} =$
$$-\frac{3}{\sqrt{66}} + \frac{2}{\sqrt{66}} + \frac{1}{\sqrt{66}} = 0$$

$$\bar{\mathbf{v}}_1 \cdot \bar{\mathbf{v}}_3 = \frac{3}{\sqrt{11}} \cdot \left(-\frac{1}{\sqrt{66}}\right) + \frac{1}{\sqrt{11}} \cdot \left(-\frac{4}{\sqrt{66}}\right) + \frac{1}{\sqrt{11}} \cdot \frac{7}{\sqrt{66}} =$$
$$= -\frac{3}{11\sqrt{6}} - \frac{4}{11\sqrt{6}} + \frac{7}{11\sqrt{6}} = 0$$

$$\bar{\mathbf{v}}_2 \cdot \bar{\mathbf{v}}_3 = -\frac{1}{\sqrt{6}} \cdot \left(-\frac{1}{\sqrt{66}}\right) + \frac{2}{\sqrt{6}} \cdot \left(-\frac{4}{\sqrt{66}}\right) + \frac{1}{\sqrt{6}} \cdot \frac{7}{\sqrt{66}} = 0$$

1) 0/g
2) unit

THEOREM 6

An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.

$$U^T \cdot U = I$$

$$U^{-1} = U^T$$

PROOF To simplify notation, we suppose that U has only three columns, each a vector in $\mathbb{R}^m$. The proof of the general case is essentially the same. Let $U = [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3]$ and compute

$$U^T U = \begin{bmatrix} \mathbf{u}_1^T & \mathbf{u}_2^T & \mathbf{u}_3^T \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1^T \mathbf{u}_1 & \mathbf{u}_1^T \mathbf{u}_2 & \mathbf{u}_1^T \mathbf{u}_3 \\ \mathbf{u}_2^T \mathbf{u}_1 & \mathbf{u}_2^T \mathbf{u}_2 & \mathbf{u}_2^T \mathbf{u}_3 \\ \mathbf{u}_3^T \mathbf{u}_1 & \mathbf{u}_3^T \mathbf{u}_2 & \mathbf{u}_3^T \mathbf{u}_3 \end{bmatrix}$$

$\mathbf{u}_1 \cdot \mathbf{u}_2 = 0$
(4)
 $\mathbf{u}_1 \cdot \mathbf{u}_1 = 1$

The entries in the matrix at the right are inner products, using transpose notation. The columns of U are orthogonal if and only if

$$\mathbf{u}_1^T \mathbf{u}_2 = \mathbf{u}_2^T \mathbf{u}_1 = 0, \quad \mathbf{u}_1^T \mathbf{u}_3 = \mathbf{u}_3^T \mathbf{u}_1 = 0, \quad \mathbf{u}_2^T \mathbf{u}_3 = \mathbf{u}_3^T \mathbf{u}_2 = 0 \quad (5)$$

The columns of U all have unit length if and only if

$$\mathbf{u}_1^T \mathbf{u}_1 = 1, \quad \mathbf{u}_2^T \mathbf{u}_2 = 1, \quad \mathbf{u}_3^T \mathbf{u}_3 = 1 \quad (6)$$

$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

The theorem follows immediately from (4)–(6). ■

THEOREM 7

Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}$ and $\mathbf{y}$ be in $\mathbb{R}^n$. Then

- $\|U\mathbf{x}\| = \|\mathbf{x}\|$
- $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$
- $(U\mathbf{x}) \cdot (U\mathbf{y}) = 0$ if and only if $\mathbf{x} \cdot \mathbf{y} = 0$

EXAMPLE 6 Let $U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix}$. Notice that U has orthonormal columns and

$$U^T U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 2/3 & -2/3 & 1/3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Verify that $\|U\mathbf{x}\| = \|\mathbf{x}\|$

$$1) \quad \|\mathbf{x}\| = \sqrt{(\sqrt{2})^2 + 3^2} = \sqrt{2+9} = \sqrt{11}$$

$$2) \quad U \cdot \mathbf{x} = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

$$3) \quad \|U \cdot \mathbf{x}\| = \sqrt{3^2 + (-1)^2 + 1^2} = \sqrt{9+1+1} = \sqrt{11}$$